

CBSE / NCERT | CLASS X | 2026-27 PRACTICE EDITION

Science

Assertion and Reason

All textbook chapters, with scope labels

16 chapters | 32 questions | answer key with explanations

How to answer

Judge Assertion (A) and Reason (R) separately, then decide whether R explains A.

A	Both A and R are true, and R explains A.
B	Both A and R are true, but R does not explain A.
C	A is true, but R is false.
D	A is false, but R is true.

The chapter set includes textbook extension chapters too. Status labels distinguish current board content, formative/project/map-only items and chapters not listed in the cited annual syllabus.

01. Chemical Reactions and Equations

Current course

Q1

Assertion (A): Burning magnesium in oxygen is a combination reaction.

Reason (R): Two reactants combine to form a single product, magnesium oxide.

Q2

Assertion (A): Iron displaces copper from copper sulphate solution.

Reason (R): Iron is more reactive than copper.

02. Acids, Bases and Salts

Current course

Q1

Assertion (A): An aqueous acid solution turns blue litmus red.

Reason (R): Acids furnish hydrogen ions in aqueous solution.

Q2

Assertion (A): Tooth decay becomes more likely when the mouth becomes too acidic.

Reason (R): Basic toothpaste can help neutralise excess acid in the mouth.

03. Metals and Non-metals

Current course

Q1

Assertion (A): Sodium is stored under kerosene.

Reason (R): Sodium reacts vigorously with oxygen and moisture in air.

Q2

Assertion (A): Ionic compounds generally conduct electricity when molten.

Reason (R): Their ions are free to move in the molten state.

04. Carbon and Its Compounds

Current course

Q1

Assertion (A): Carbon forms a very large number of compounds.

Reason (R): Carbon atoms can form four covalent bonds and link to one another in chains.

Q2**Assertion (A):** Soap forms scum more readily in hard water than in soft water.**Reason (R):** Calcium and magnesium ions react with soap to form insoluble salts.

05. Periodic Classification of Elements

Textbook extension; not listed in the 2025-26 annual Science syllabus

Q1**Assertion (A):** In the modern periodic table, elements are arranged by increasing atomic number.**Reason (R):** The modern periodic law relates properties to atomic number.**Q2****Assertion (A):** Elements in the same group often have similar chemical properties.**Reason (R):** They have the same number of electrons in their outermost shell.

06. Life Processes

Current course

Q1**Assertion (A):** Plants need stomata for gas exchange.**Reason (R):** Stomata are pores that can open and close on leaf surfaces.**Q2****Assertion (A):** The left ventricle pumps oxygenated blood to the body.**Reason (R):** Its thick muscular wall generates pressure for systemic circulation.

07. Control and Coordination

Current course

Q1**Assertion (A):** A reflex action can occur before a person consciously thinks about it.**Reason (R):** A reflex arc can relay a signal through the spinal cord.**Q2****Assertion (A):** A shoot usually bends towards light.**Reason (R):** Auxin can accumulate on the shaded side and promote greater cell elongation there.

08. How Do Organisms Reproduce?

Current course

Q1**Assertion (A):** Binary fission can produce two daughter cells from one parent cell.**Reason (R):** The parent cell divides after its genetic material is copied.**Q2****Assertion (A):** Fertilisation restores the diploid chromosome number in humans.**Reason (R):** A haploid sperm and a haploid egg fuse to form a zygote.

09. Heredity and Evolution

Current course

Q1**Assertion (A):** A child receives genetic information from both parents.**Reason (R):** A human egg and sperm each carry one set of chromosomes.**Q2****Assertion (A):** In humans, the father's sperm determines whether the child is genetically XX or XY.**Reason (R):** Egg cells carry an X chromosome, while sperm cells may carry X or Y.

10. Light - Reflection and Refraction

Current course

Q1**Assertion (A):** A concave mirror can form a real image of an object beyond its focus.**Reason (R):** Reflected rays from such an object can actually converge in front of the mirror.**Q2****Assertion (A):** A convex lens has positive power when its focal length is measured in metres.**Reason (R):** Lens power in dioptres is the reciprocal of focal length in metres.

11. The Human Eye and the Colourful World

Current course

Q1**Assertion (A):** A myopic person may see distant objects clearly with a concave lens.**Reason (R):** A concave lens diverges incoming light before it reaches the eye.

Q2

Assertion (A): The sky appears blue in daytime because shorter visible wavelengths are scattered more by air molecules.

Reason (R): Blue light has a shorter wavelength than red light.

12. Electricity

Current course

Q1

Assertion (A): The current through a resistor doubles when its voltage doubles, if its resistance stays constant.

Reason (R): Ohm's law gives $V = IR$.

Q2

Assertion (A): Two identical resistors in parallel have less equivalent resistance than either resistor alone.

Reason (R): Parallel branches provide additional paths for charge flow.

13. Magnetic Effects of Electric Current

Current course

Q1

Assertion (A): A current-carrying wire produces a magnetic field around it.

Reason (R): A compass near the wire can deflect when current flows.

Q2

Assertion (A): The direction of force on a current-carrying conductor in a magnetic field can be found with Fleming's left-hand rule.

Reason (R): The rule relates the directions of magnetic field, current and force.

14. Sources of Energy

Textbook extension; not listed in the 2025-26 annual Science syllabus

Q1

Assertion (A): Solar energy is a renewable source on a human timescale.

Reason (R): The Sun continuously supplies energy to Earth over the timescale of human use.

Q2

Assertion (A): A good source of energy should release a large amount of energy per unit mass.

Reason (R): A high calorific value means more energy is obtained from a unit mass of fuel.

15. Our Environment

Current course

Q1**Assertion (A):** Energy available to organisms decreases at successive trophic levels.**Reason (R):** Only a fraction of energy is passed to the next trophic level.**Q2****Assertion (A):** Non-biodegradable substances can persist in an ecosystem for a long time.**Reason (R):** Decomposers cannot readily break down many such materials.

16. Management of Natural Resources

Textbook extension; not listed in the 2025-26 annual Science syllabus

Q1**Assertion (A):** Sustainable resource management considers the needs of future generations.**Reason (R):** Resources may be limited and can be depleted by excessive use.**Q2****Assertion (A):** Large dams can create both benefits and social or environmental costs.**Reason (R):** Reservoirs may displace communities and alter river ecosystems.

Answer key and explanations

01. Chemical Reactions and Equations

Q1 - A. The reaction forms one product from magnesium and oxygen.

Q2 - A. A more reactive metal can displace a less reactive metal from its salt solution.

02. Acids, Bases and Salts

Q1 - A. The hydrogen-ion behaviour is associated with the characteristic litmus change.

Q2 - A. Neutralising acid helps protect tooth enamel from acid attack.

03. Metals and Non-metals

Q1 - A. Kerosene keeps air and water away from the reactive metal.

Q2 - A. Mobile ions carry electric charge through the molten compound.

04. Carbon and Its Compounds

Q1 - A. Tetravalency and catenation together allow many stable structures.

Q2 - A. Those insoluble salts are the scum seen in hard water.

05. Periodic Classification of Elements

Q1 - A. Atomic number is the basis of the modern arrangement and periodic law.

Q2 - A. Similar valence-electron configurations lead to similar chemical behaviour.

06. Life Processes

Q1 - A. Opening and closing these pores regulates gas exchange and water loss.

Q2 - A. The strong contraction sends blood through the aorta to body tissues.

07. Control and Coordination

Q1 - A. The rapid spinal pathway produces a response before conscious processing is complete.

Q2 - A. Unequal growth on the shaded side bends the shoot toward the light.

08. How Do Organisms Reproduce?

Q1 - A. One cell separates into two cells carrying copied genetic material.

Q2 - A. The two haploid sets combine in the zygote.

09. Heredity and Evolution

Q1 - A. The two gametes contribute one set each at fertilisation.

Q2 - A. The sperm's X or Y chromosome completes the child's sex-chromosome pair.

10. Light - Reflection and Refraction

Q1 - A. Actual convergence of reflected rays forms a real image.

Q2 - A. A convex lens has positive focal length, so its reciprocal is positive.

11. The Human Eye and the Colourful World

Q1 - A. The diverging lens moves the image of a distant object back onto the retina.

Q2 - A. Shorter wavelengths undergo stronger scattering in the atmosphere.

12. Electricity

Q1 - A. With fixed R , current is directly proportional to voltage.

Q2 - A. Adding parallel paths reduces the total opposition to current.

13. Magnetic Effects of Electric Current

Q1 - A. The compass deflection is evidence of the field produced by current.

Q2 - A. Its three mutually perpendicular fingers represent those directions.

14. Sources of Energy

Q1 - A. Its replenishment makes it renewable for human planning.

Q2 - A. High energy density is one useful property of a fuel.

15. Our Environment

Q1 - A. Energy is lost through life processes and heat at each transfer.

Q2 - A. Slow breakdown allows the substances to accumulate in the environment.

16. Management of Natural Resources

Q1 - A. Using resources carefully helps preserve availability over time.

Q2 - B. Both statements are true, but the reason describes costs rather than explaining all benefits and costs.

Scope and sources

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Question text in this workbook is original practice material. Chapter labels and scope notes are provided for navigation; check your school's current list before using extension chapters for exam revision.

CBSE Class X Science curriculum 2025-26

https://cbseacademic.nic.in/web_material/CurriculumMain26/Sec/Science_Sec_2025-26.pdf

CBSE curriculum portal

https://cbseacademic.nic.in/curriculum_2027.html

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